

Statistical and Computational Models

Lecture 3: Nonparametric regression

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Roadmap

Beyond a fixed form

One tuning parameter

Training and test MSE

Parametric vs. Non-Parametric Models

PARAMETRIC

Assume a form: $f(\mathbf{x}) = \beta_0 + \beta_1 \mathbf{x}$.

Estimate $p + 1$ parameters.

NON-PARAMETRIC

Assume only smoothness.

$\hat{f}(\mathbf{x})$ is a local average of nearby y .

Using a parametric model cost us **approximation error**: if our function is a poor approximation of the DGP, no amount of data will rescue us.

Non-parametric models can reduce approximation error, but like in life there is **no free lunch...**

Local averaging

Recall we want $f(x_0)$, turnout among observations of age **exactly** x_0 :

$$f(x_0) = \mathbb{E}[Y \mid X = x_0]$$

Problem: if X is continuous there are scarce observations with $x_i = x_0$.

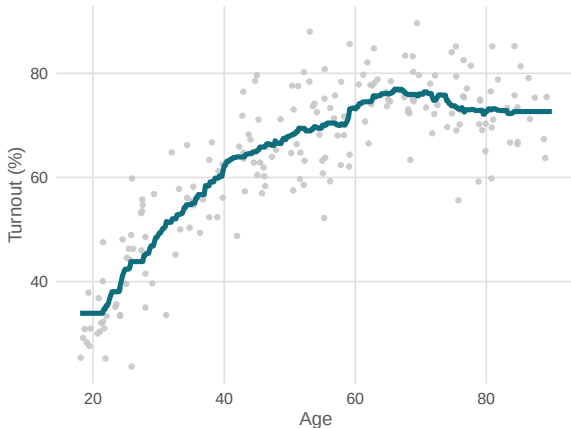
Let $A_k(x_0)$ hold the k **nearest neighbours** of x_0 in X :

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{x_i \in A_k(x_0)} y_i$$

Read: the estimate at x_0 is the mean turnout inside $A_k(x_0)$.

There is no β to estimate, but we have to pick k . How?

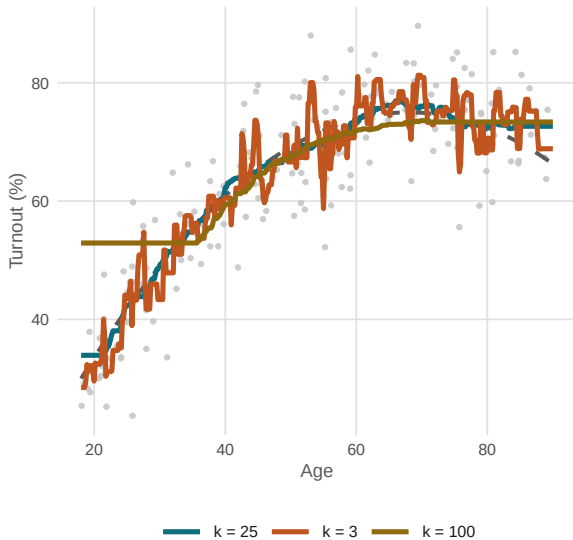
Building the k-nn regression line



Pick a k (e.g. 25); for each age average the closest k observations.

Do that at every age and you get back a line. But **why $k = 25$?**

k -NN on turnout



Interpretation

k -NN has no $\hat{\beta}$, so there is nothing obvious to report in a table.

However, we can **evaluate the function at discrete points**, and draw contrasts.

For a one unit change in X , k -nn gives us **+1.27** at age 30, **+0.61** at 60, **−0.11** at 80.

A linear regression gives us a neater summary: **+0.60** points a year, homogeneous across all observations and the span of X .

Note that uncertainty will not have a closed-form solution, so **resample**, e.g. a bootstrap.

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What k controls

k is what is called a **tuning parameter**; in this case it governs how much to **smooth** our function.

At $k = 1$ we do no smoothing, while at $k = n$ the fit is a single horizontal line at $\bar{y}$ – that is the linear regression with only an intercept! Generally:

SMALL k

Few neighbours.

The fit follows the sample.

High **variance**.

LARGE k

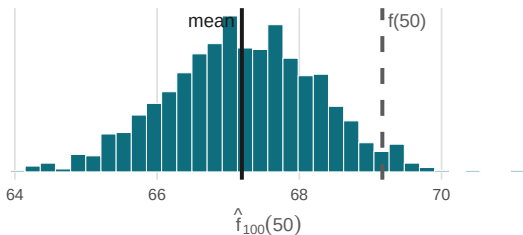
Many neighbours.

We smooth between ages.

High **bias**.

Bias and variance

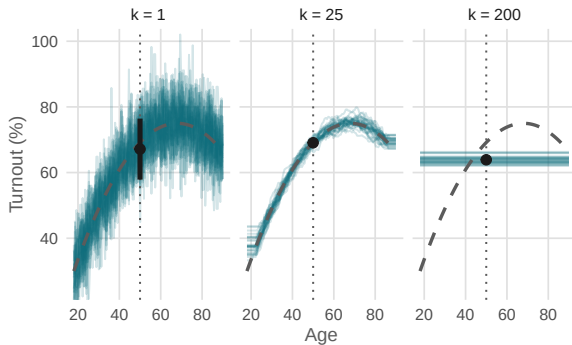
Suppose we have a proposed estimator $\hat{f}(x_0)$. Draw a sample, compute $\hat{f}(x_0)$. Draw another, compute it again. Over repeated samples $\hat{f}(x_0)$ has a **sampling distribution**.



$$\text{Bias}(x_0) = \mathbb{E}[\hat{f}(x_0)] - f(x_0), \quad \text{Var}(x_0) = \mathbb{E}\left[\left(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2\right]$$

Read: bias is the **central tendency** of that distribution, against the estimand. Variance is its **spread**.

Bias and variance



Consider 30 samples, the same estimators refitted on each.

At $k = 1$ the curves scatter around the true f . At $k = 200$ they agree with each other and all miss it. The bar is ± 1 sd of $\hat{f}(50)$.

Bias and variance, in k

$$\mathbb{E}[\text{EPE}(\mathbf{x}_0)] = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\left[f(\mathbf{x}_0) - \frac{1}{k} \sum_{\ell=1}^k f(\mathbf{x}_{(\ell)}) \right]^2}_{\text{squared bias}} + \underbrace{\frac{\sigma^2}{k}}_{\text{variance}}$$

Note that k sits in **two** of the three terms, but cannot help with irreducible error.

Variance is σ^2/k : we average k independent noise terms, so their variance divides by k .

Bias is the gap between $f(\mathbf{x}_0)$ and the average of f over the neighbourhood. Widen the neighbourhood and it grows – we are using poorer stand-ins for the truth.

So which do we want, **big k or small k** ? In a moment...

Step functions

First let's take a detour and look at an alternative non-parametric estimator:

Cut age into **bins** and fit a **constant** in each:

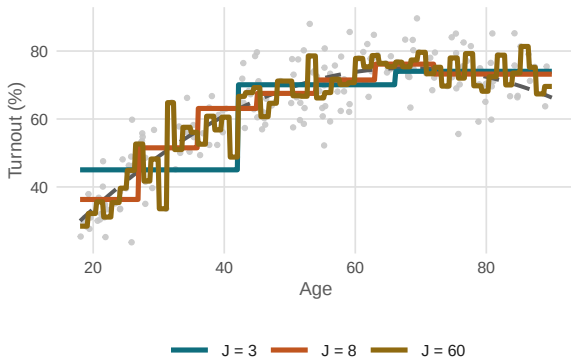
$$\hat{f}(x) = \sum_{j=1}^J \hat{c}_j \mathbf{1}\{x \in B_j\}$$

Read: a local level (mean), with the bins fixed in advance.

The number of bins J is the tuning parameter. How do we pick J ?

Question: how is this different from k -NN? k -NN's bin moves with $\mathbf{x}_0$.

Step functions



$J = 8$ holds about 25 observations a bin: the $k = 25$ neighbourhood, on a grid.

Move the bin with $\mathbf{x}_0$ and it is a **local smoother**. Weight by distance or a **kernel** (we will see this in a later lecture).

Back to the linear model

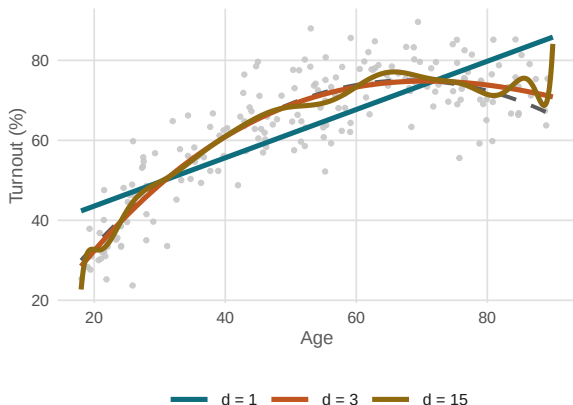
One more detour. What if we propose a linear regression but make its **order** the tuning parameter.

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \varepsilon_i$$

Read: least squares, linear in β , but with as many columns and parameters as $d + 1$.

$d = 1$ is a straight line, $d = 3$ the cubic form, and so on.

Back to the linear model



Is the **wiggliest one the best**? It turns out this is the same problem as picking k , or picking J .

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Training vs. test vs. EPE

Recall that we want the **EPE**, so we can pick $\hat{f}(\cdot)$ that minimizes the the average loss over the whole DGP.

We cannot see the EPE, so we need an empirical alternative:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(x_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(X))]$	the population (or very many draws from the DGP)	never

Two errors

$$\text{MSE}_{\text{train}} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}(x_i) \right)^2, \quad \text{MSE}_{\text{test}} = \frac{1}{m} \sum_{j=1}^m \left(y_j - \hat{f}(x_j) \right)^2$$

Read: the second sum runs over observations $\hat{f}$ has not been fit to.

Note: we are using squared loss, hence we call these $\text{MSE}_{\text{train}}$ and MSE_{test} .

Check: which of these can you get from `summary(lm())`? The first.
Always!

Why training MSE lies

Let $f^\dagger$ minimise EPE over the class, and $\hat{f}$ minimise $\text{MSE}_{\text{train}}$.

$\hat{f}$ was chosen to minimise $\text{MSE}_{\text{train}}$ for the realised sample. So $\text{MSE}_{\text{train}}(\hat{f}) \leq \text{MSE}_{\text{train}}(f^\dagger)$.

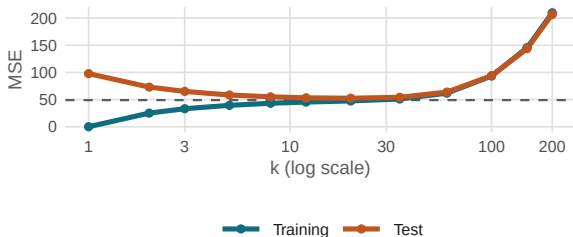
But $\mathbb{E}[\text{MSE}_{\text{train}}(f^\dagger)] = \text{EPE}(f^\dagger)$.

$$\mathbb{E}[\text{MSE}_{\text{train}}(\hat{f})] \leq \text{EPE}(f^\dagger) \leq \mathbb{E}[\text{EPE}(\hat{f})]$$

QED

At $k = 1$ the fit runs through every observation: $\text{MSE}_{\text{train}} = 0$ (a perfect model!).

Train and test, compared



Let's see this in action. Over 40 samples, calculate both the test and the train curves.

$\hat{f}$ **overfits** when some less flexible fit would give lower test MSE.

So, how do we pick k or J or d ? We'll learn that next week.

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **3.5, 7.1–7.2**. statlearning.com.
- Hastie, T., Tibshirani, R. & Friedman, J. (2009) *The Elements of Statistical Learning*, 2nd ed. – 2.3, 7.1–7.3.